

## Tutorial N° 2: Process Management

**Objectives:** By the end of this series, you will be able to:

- Understand and master the operating principle of scheduling policy.
- Know how to draw a Gantt chart and manage different queues, especially multi-level queues.
- Apply performance criteria to each type of system studied in the series.

### Exercise 1: Basic Scheduling Algorithms

Consider five processes P1, P2, P3, P4, and P5 that arrive in the system (ready queue) at the same time (at time 0) and whose characteristics are summarized in the following table (the smaller priority number indicates higher priority).

| Process | Order of processes in the Ready queue | Execution Time | Priority |
|---------|---------------------------------------|----------------|----------|
| P1      | 1                                     | 2              | 2        |
| P2      | 2                                     | 6              | 4        |
| P3      | 3                                     | 10             | 3        |
| P4      | 4                                     | 4              | 5        |
| P5      | 5                                     | 12             | 1        |

For each of the following scheduling algorithms: **FCFS**, **SJF** (Shortest Job First), **Fixed Priority** and **Round Robin** (with a quantum = 2).

1. Draw the Gantt chart.
2. Calculate :
  - a. The Average Turnaround Time (ATT)
  - b. The Average Waiting Time (AWT)
  - c. The Average Response Time (ART)

### Exercise 2: Scheduling with Multiple I/O Devices

Consider four processes P1, P2, P3, and P4, which perform calculations on the CPU and request I/O operations (see the table below):

| Process | Arrival Time | Execution Time                        |
|---------|--------------|---------------------------------------|
| P1      | 0            | 7 CPU units, 3 I/O units, 5 CPU units |
| P2      | 1            | 6 CPU units, 4 I/O units, 4 CPU units |
| P3      | 9            | 5 CPU units                           |
| P4      | 12           | 1 CPU unit, 4 I/O units, 2 CPU units  |

**Explanation:** The first line means that process P1 arrives in the system at time 0, and its execution requires, in order: 7 CPU units, 3 I/O units, and 5 CPU units, and so on for the other processes.

- Draw the execution diagram in each of the following cases, showing the state of the two queues (Ready and I/O):
  - 1) All processes use the same I/O device, and the scheduler uses the FCFS algorithm.
  - 2) All processes use the same I/O device, and the scheduler uses the Round Robin algorithm with a quantum of 5 units. In this case, calculate the average waiting time and the average response time.
  - 3) Each process has its own I/O device, and the scheduler uses the Round Robin algorithm with a quantum of 5 units. In this case, calculate the average waiting time and average response time.

### Exercise 3: Preemptive and Non-Preemptive Priority Scheduling

Consider five processes P1, P2, P3, P4, and P5, whose characteristics are summarized in the following table (the smaller priority number indicates higher priority).

| Process | Arrival Time<br>(in Ready Queue) | Execution Time | Priority |
|---------|----------------------------------|----------------|----------|
| P1      | 2                                | 2              | 1        |
| P2      | 4                                | 6              | 4        |
| P3      | 5                                | 3              | 2        |
| P4      | 0                                | 4              | 5        |
| P5      | 0                                | 7              | 3        |

- For both types of scheduling: Non-Preemptive-Priority Scheduling and Preemptive-Priority Scheduling.
  1. Draw the Gantt chart, showing the state of the Ready Queue.
  2. Calculate
    - a. The Average Turnaround Time.
    - b. The Average Waiting Time.
    - c. The Average Response Time.

### Exercise 4 : Multiprocessor Scheduling

Consider an operating system that schedules processes using the Round Robin algorithm. Suppose this system has two CPUs, CPU1 and CPU2, and a single I/O unit.

All ready processes are in the same Ready Queue, managed by a single scheduler that applies the Round Robin algorithm with a quantum ( $Q_t = 3$ ).

- Consider the three processes P1, P2, and P3 described in the following table:

| Process | Arrival Time | Execution Time                        |
|---------|--------------|---------------------------------------|
| P1      | 0            | 4 CPU units, 2 I/O units, 2 CPU units |
| P2      | 2            | 3 CPU units, 4 I/O units, 2 CPU units |
| P3      | 3.5          | 5 CPU units                           |

Initially, process P1 is selected by the scheduler to be executed by CPU1. If multiple events occur simultaneously, we assume the following priorities:

- CPU1 has priority over CPU2 for executing ready processes (i.e., the Dispatcher directs processes to CPU1 first).
  - At the end of a quantum, the current process is suspended only if the Ready Queue is not empty (i.e., as long as there are no waiting processes; in this case, the current process may receive a second quantum to continue execution).
  - The processing at the end of a quantum (clock interrupt) has higher priority than the end of I/O, this last has a higher priority than the arrival of new processes in the system.
1. In such system, explain the role of the Dispatcher.
  2. Draw the Gantt charts showing the allocation of CPU1, CPU2, and the I/O unit.
  3. Repeat the task by drawing a two-dimensional execution graph.
  4. Calculate the average turnaround time, response time, and waiting time.

### Exercise 5: SRTF Algorithm

Assuming that each process has its own I/O device, apply the SRTF scheduling policy (Shortest Remaining Time First) to the set of processes described in the following table:

| Process | Arrival Time | Execution Time in ms                                    |
|---------|--------------|---------------------------------------------------------|
| A       | 00           | 100 ms CPU                                              |
| B       | 10           | 28 ms CPU, 150 ms I/O, 75 ms CPU, 160 ms I/O, 20 ms CPU |
| C       | 50           | 30 ms CPU                                               |
| D       | 12           | 4 ms CPU, 200 ms I/O, 4 ms CPU                          |
| E       | 30           | 30 ms CPU                                               |

- 1) By applying SRTF policy. Draw a bi-dimensional graph of the execution flow. You must show on the graph: the arrival time, the burst time, the remaining execution time at each state change, and the completion time. What is the average turnaround time, response time and waiting time?
- 2) What is the average turnaround time if the same processes are scheduled using the Round Robin policy with a quantum= 5 units?
- 3) Compare the results obtained by applying SRTF and Round Robin and draw your conclusions.

### Exercise 6 : Multi-Level Queue Without Feedback

Consider an operating system that uses a multi-level queue without feedback. The queue is divided into two levels:

- **Level 1 (Foreground)** : Uses the Round Robin algorithm with a time quantum = 10 ms..
- **Level 2 (Background)** : Uses the First Come First Served (FCFS) algorithm..

The process classification follows these rules:

- Interactive processes (Foreground) have a higher priority than batch processes running in the background (Background).
- A process cannot move between levels.

The processes to be scheduled are as follows:

| Process | Type       | Arrival Time | Execution Schema (CPU & I/O)   |
|---------|------------|--------------|--------------------------------|
| P1      | Foreground | 0 ms         | 10 ms CPU, 9 ms E/S, 10 ms CPU |
| P2      | Background | 5 ms         | 20 ms CPU                      |
| P3      | Foreground | 10 ms        | 15 ms CPU, 10 ms E/S, 5 ms CPU |
| P4      | Background | 15 ms        | 15 ms CPU                      |

1. Draw the **Gantt chart** showing the execution of processes for both levels (Foreground and Background).
2. Calculate:
  - The **average turnaround time**.
  - The **average waiting time**.
  - The **average response time** for interactive processes.
3. Analyze the **system performance** by comparing the waiting and turnaround times of **Foreground and Background** processes.

#### Exercise 7 : Multi-Level Queue Without Feedback

Consider an operating system that uses a **multi-level queue without feedback**. The queue is divided into three levels:

- **Level 1 (Real-Time - High Priority)**: Uses the **Round Robin** algorithm with a **time quantum = 4 ms**.
- **Level 2 (Foreground - Medium Priority)**: Uses the **Shortest Job First (SJF)** algorithm.
- **Level 3 (Background - Low Priority)**: Uses the **First Come First Served (FCFS)** algorithm.

Process classification follows these rules:

- **Real-time processes** have the highest priority.
- **Foreground processes** have medium priority.
- **Background (batch) processes** have the lowest priority.
- **A process cannot move between levels.**

The processes to be scheduled are:

| Process | Type       | Arrival Time | CPU Execution Time |
|---------|------------|--------------|--------------------|
| P1      | Real-Time  | 0 ms         | 12 ms              |
| P2      | Foreground | 1 ms         | 8 ms               |
| P3      | Real-Time  | 2 ms         | 4 ms               |
| P4      | Background | 3 ms         | 10 ms              |
| P5      | Foreground | 5 ms         | 6 ms               |

1. Draw the **Gantt chart** respecting the queue scheduling policy.
2. Calculate the **average waiting time** and **average turnaround time** for all processes.
3. Explain the **impact of this scheduling policy** on **system performance** and the **fairness between processes**.

### Exercise 8 : Multi-Level Queue Without Feedback

Consider an operating system that uses a multi-level queue without feedback. The queue is divided into two levels:

- **Level 1 (Foreground)** : Uses the **Round Robin algorithm with a time quantum = 20 ms**.
- **Level 2 (Background)** : Uses the **First Come First Served (FCFS) algorithm**.

1. Construct the **state diagram** corresponding to the defined execution scheme. Explain your choices of states and transitions.
2. The following **five processes** are scheduled according to the defined scheduling policy:

| Process | Type       | Arrival Time | Execution Schema (CPU & I/O)                            |
|---------|------------|--------------|---------------------------------------------------------|
| A       | Background | 00           | 130 ms CPU                                              |
| B       | Foreground | 10           | 20 ms CPU, 100 ms E/S, 70 ms CPU, 100 ms E/S, 20 ms CPU |
| C       | Foreground | 20           | 30 ms CPU                                               |
| D       | Background | 30           | 10 ms CPU, 100 ms E/S, 10 ms CPU                        |
| E       | Background | 01           | 100 ms CPU                                              |

- Draw the **execution graph** for the five processes. For each process, specify:
  - The **start and end execution times**.
  - The **remaining execution time** at each state change.

### Exercise 9 : Multi-Level Queue with Feedback

Consider an operating system that uses a **multi-level queue with feedback**. The queue is divided into three levels:

- **Level 1:** Uses the **Round Robin algorithm with a time quantum = 8 ms**.
- **Level 2:** Uses the **Round Robin algorithm with a time quantum = 16 ms**.
- **Level 3:** Uses the **FCFS algorithm**.

The **feedback rules** are as follows:

- If a process **does not complete within Level 1**, it is **demoted to Level 2**.
- If a process **does not complete within Level 2**, it is **demoted to Level 3**.
- **Processes cannot move back to higher levels**.

The processes to be scheduled are:

| Process | Arrival Time | Execution Schema (CPU & I/O) |
|---------|--------------|------------------------------|
| P1      | 0 ms         | 8 ms CPU, 8 ms E/S, 8 ms CPU |
| P2      | 5 ms         | 16 ms CPU                    |
| P3      | 10 ms        | 16 ms CPU, 16 ms E/S         |
| P4      | 15 ms        | 12 ms CPU                    |

1. Draw the **Gantt chart** showing the execution of processes for the three levels.
2. Calculate:
  - The **average turnaround time**.
  - The **average waiting time**.
  - The **average response time** for these processes.

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3. Analyze the **impact of feedback** on system performance. Compare the results with a **non-feedback system**.

**Exercise 10 : Multi-Level Queue with Feedback**

Consider an operating system that uses a **multi-level queue with feedback**. The queue is divided into three levels:

- **Level 1:** Uses the **Round Robin algorithm with a time quantum of 3 ms**.
- **Level 2:** Uses the **Round Robin algorithm with a time quantum of 6 ms**.
- **Level 3:** Uses the **FCFS algorithm**.

The **feedback rules** are as follows:

- If a process **does not complete within Level 1**, it is **demoted to Level 2**.
- If a process **does not complete within Level 2**, it is **demoted to Level 3**.
- **Processes cannot move back to higher levels**.

The processes to be scheduled are:

| Process | Arrival Time | CPU Execution Time |
|---------|--------------|--------------------|
| P1      | 0 ms         | 14 ms              |
| P2      | 2 ms         | 10 ms              |
| P3      | 4 ms         | 5 ms               |

1. Draw the **Gantt chart** showing the execution of processes for the three levels.
2. Calculate the **average waiting time** and **average turnaround time** for these processes.
3. Compare the results with a **classic Round Robin scheduling** and explain the **advantages/disadvantages** of this type of queue.